

Semantic Segmentation Model of MSI/Sentinel2 dataset CerraData4m Using Efficient-Unet and U-net

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Abstract—This study adapted Efficient-Unet model to satellite imagery from MultiSpectral Imager (MSI) sensor onboard Sentinel-2 constellation, comparing the same model with and without pretrained weights. The used dataset was Cerradata4mm, based on TerraClass project. This study proved that using pretrained weights performs better, generally, than without using them. Another results was the variation of imbalanced classes in the metrics in the models.

Index Terms—remote sensing, deep learning, convolutional neural networks, imbalanced classes

I. INTRODUCTION

Semantic segmentation has been widely used for remote sensing, once remote sensing is a field with numerous amount of studies and necessity of understanding the Earth in different ways and in different periods of time. One of the studies is Land Use and Land Cover (LULC), which generates general understanding of the area of study in order to study change rates, such as deforestation, urban growing and so on [1]. Aligned with it, nowadays, there are several models that use Convolution Neural Networks (CNN) to solve semantic segmentation, providing a more accurate and faster model by the day.

Using semantic segmentation algorithms for remotely sensed data is common use in academia, with different focuses. However, models have not yet been able to provide same accuracy as human classification (handcrafted) and, for this reason, projects such as *Programa de Monitoramento da Floresta Amazônica Brasileira por Satélite* (PRODES), *Sistema de Detecção de Desmatamento em Tempo Real* (DETER) as well as *sistema de mapeamento do uso e ocupação da terra*, TerraClass [2], [3] still use human interpreters for assure these detections accuracies.

Given this problem, researchers are constantly working on providing trustworthy datasets and labelling for this type of data, considering the impacts a low accuracy model can have. One solution for studying LULC of Cerrado Biome was the publication of Cerradata4mm dataset [4]. From this dataset, the aim of this study is to run a CNN model in order to achieve

best possible accuracy with the expectation of aiding fellow researchers to interpretate this type of dataset.

The paper is organised as follows, we first explain the dataset in Section II, then we explain the model we were based on (Section IV), followed by the explanation of the model adaptations to the dataset IV-C. After that, we explain how the experiments were conducted in Section V, showing the results in Section VI and concluding our study in Section VII.

II. DATASET

A. Imagery

The used dataset is entitled Cerradata4mm [4], a dataset created with images from the MultiSpectral Imager (MSI) sensor onboard Sentinel-2 mission [5] and Sentinel-1 SAR C-band mission [6]. The Sentinel-2/MSI mission was launched by the European Space Agency (ESA). This mission has twin satellites and can sense the Earth with a 5-day revisit time and spatial resolution varying from 10m to 60m, depending on the channel. It has 12 spectral channels, all used for this dataset. For more information on the satellite, please refer to [5].

As for the latter, it is also from ESA and it was first launched in 2014, with twin satellites using Synthetic Aperture Radar (SAR) C-band (5.405 GHz or $\approx 5.5\text{cm}$), it has two main polarisations: VV and VH, which V stands for vertical and H is horizontal. This mission has a 6-day revisit time with resolution varying according to incidence angle.

For the area, the dataset was based on the Cerrado biome, which is a savanna ecoregion, specifically *Bico do Papagaio* region, as shown in Fig. 1. According to Miranda et al (2025) [4], the dataset is a mosaic of 18 satellite tiles (each 1098x1098 pixels), resulting in 30,291 images of 128×128 pixels.

B. Classes

The classes used for this dataset are for Land Use and Land Cover (LULC), and there are two possibilities: 7 and 14 classes, shown in Tab. I. These classes were based on the TerraClass project [2], from 2022 map.

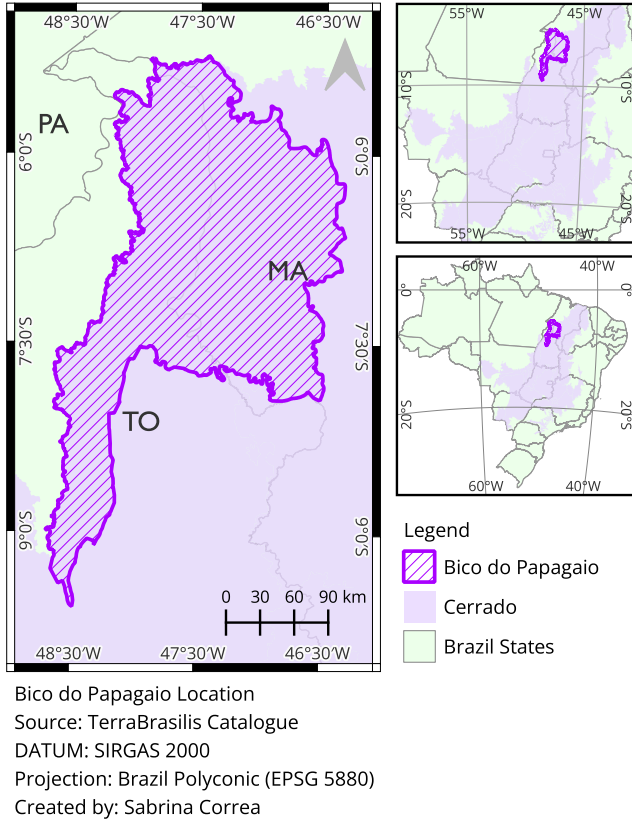


Fig. 1. Area of study: Bico do Papagaio. In this image, the hatched area refers to the study area, which is within the States of Maranhão (MA) and Tocantins (TO), being located on the northeast of Brazil. Adapted from: Miranda et al (2025) [4], TerraBrasilis [7].

TABLE I
LAND COVER CLASSIFICATION SUMMARY

L1 Class	L1 %	L2 Class	L2 %	Samples
Pasture (Pa)	35.25	Pasture (Pa)	35.25	174,929,011
Arboreal (Ab)	58.43	Primary Natural Vegetation (V1)	52.21	259,126,072
		Secondary Natural Vegetation (V2)	6.22	30,874,079
Agriculture (Ag)	4.43	Perennial (Pr)	0.07	371,519
		Semi-perennial (SP)	0.15	767,292
		Temp. Agriculture 1 cycle (T1)	0.27	1,348,807
		Temp. Agriculture 1+ cycles (T1+)	1.15	5,702,780
		Forestry (Ft)	1.49	7,388,637
		Deforestation (Df)	1.30	6,466,058
Mining (Mg)	0.01	Mining (Mg)	0.01	63,964
Building (Bg)	0.55	Urban Area (UA)	0.45	2,241,073
		Other Built Area (OB)	0.10	508,261
Water Body (Wt)	1.32	Water Body (Wt)	1.32	6,447,232
Other Uses (OU)	0.01	Other Uses (OU)	0.01	52,959

Source: Miranda et al., 2025 [4]

III. FEATURE EXTRACTION AND SELECTION

From the 14 channels available in the dataset (12 channels optical and 2 SAR), we only used 4, being: Blue, Green, Red and Near-infrared (NIR). There are two reasons for that: (i) these channels have 10m spectral resolution rather than 20 or 60m from the others, so this way, we guarantee functionality of all convolutions in the model, and (ii) the SAR bands were not used because they follow a different distribution, with different mean and average than the optical data.

Besides these four channels, we have decided to add 3 other channels, widely known in Remote Sensing for aiding LULC classes separability, being: Normalised Differentiated Vegetation Index (NDVI), Enhanced Vegetation Index 2 (EVI2) and Soil Adjusted Vegetation Index (SAVI) [1]. Their equations are presented in Equations 1, 2 and 3. In these equations, the gain factor (G) is $G = 2.5$, while the canopy background adjustments for each sensor are $L_{EVI} = 2.4$ and $L_{SAVI} = 0.5$.

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

$$EVI2 = G \cdot \frac{NIR - Red}{L_{EVI} + NIR + C \cdot Red + 10^{-6}} \quad (2)$$

$$SAVI = (1 + L_{SAVI}) \cdot \frac{NIR - Red}{NIR + Red + L_{SAVI} + 10^{-6}} \quad (3)$$

To conclude, this model has 7 channels as input: reflectance values from Blue, Green, Red and NIR, as well as remote sensing indexes NDVI, EVI2, and SAVI.

IV. MODEL

The used model is known as Efficient-Unet, proposed by Baheti et al (2020) [8], which is a fusion between U-Net [9] and EfficientNet [10]. This is an encoder-decoder structure (Fig. 2), which uses EfficientNet as encoder, followed by a bottleneck and then uses U-Net as decoder.

This structure (encoder-decoder) can be understood as an extractor of features by progressively downsampling the input image, hence reducing the feature map resolution; this results in features that extract more details from the images [8], [11]. On the other hand, the decoder part is resumed by upconvolutions from the feature map in order to recover the original image spatial information [12].

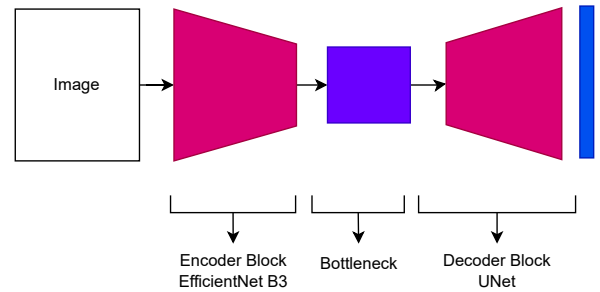


Fig. 2. General design of an encoder-decoder structure for a semantic segmentation architecture based on CNN models. Adapted from Baheti et al (2020) [8].

A. Encoder - EfficientNet

The EfficientNet model was proposed by Tan and Le in 2019 [10], which proposed a method of scaling depth models, as the ResNet Model [13]. From this, the authors created a baseline entitled EfficientNetB0, and it can be scaled up in

eight variants, up to EfficientNetB7, stating it can improve the model performance systematically by creating a balance of the coefficients and architecture (width, depth, image resolutions). This model uses the basic building block, MBConv (mobile inverted bottleneck convolution) [14]

B. Decoder - U-Net

This U-shaped architecture was originally developed for biomedicine and then adapted for other fields; it is divided into contraction and expansion parts [9], later understood as encoder and decoder. The expansion part combines information from the feature map and spatial information with a sequence of convolutions and concatenations

From this, the original EfficientNet architecture is presented in Fig. 3, which shows 7 blocks, with each block containing an MBConv.

C. Model Adaptation

Considering the input image of the original Efficient-UNet, which is $320 \times 224 \times 3$ and the used dataset, plus feature extraction and selection, resulting in an input data with size $128 \times 128 \times n$, n being the number of input features, the model needed to be adapted. The resulting general architecture is presented in Fig. 4.

Even though the input channels are different, the pretrained weights from ImageNet [15] were used. Another adaptation was that the original Efficient-Unet Model was created with EfficientNetb5 and b7 [8]; nevertheless, due to computer limitations, a smaller model, b3, was used. The activation function was LeakyReLU [16].

The used optimiser was AdamW, which weights imbalanced classes, aiming to penalise less frequent classes [17]. As for the losses, the chosen one was Focal Tversky Loss, a region-based loss proven to perform well compared to other losses for semantic segmentation [18], [19]. In the end, the Efficient-Unet model returned 11,849,039 trainable parameters.

V. EXPERIMENTAL SETUP

Using the Dataset and the model, two setups were conducted: with and without pertaining ¹ for comparison. As for the hyperparameters, they were fine-tuned to be a batch size of 64, 100 epochs, learning rate (lr) of 10^{-4} and weight decay of 10^{-3} . The entire dataset was used, and it was split into 80%/20% training and validation, respectively. All images were normalised from 0 to 1, assuming a normal distribution, which is inherent from optical satellite data [1].

The setups run on a local machine, using the graphic card NVIDIA RTX 2000 Ada Generation with 8gb, and a processor 13th Gen Intel® Core™ i7-13800H, taking about 2min30s per epoch without pretrained weights and about 6min using pretrained weights.

Considering the Early Stopping, we considered the best model as the one with the smallest combined loss and patience of 20, i.e., if the model does not show any improvements in loss for 20 epochs, it will stop. In order to increase the

possibility of data variation, data augmentation was used; those being vertical and horizontal flips, which consider real possibilities of real scenarios for remotely sensed data.

A. Evaluation Metrics

Apart from the losses, shown in Section IV-C, the model assessment can be done by accuracy metrics; the chosen ones were Overall Accuracy (OA) and Jaccard (also known as mean intersection over union, mIoU), as well as completeness metrics, Precision, Recall, Specificity and F1-score, for each class. All of them were computed from the confusion matrix (Tab.II) for each epoch. F1-score is generally used for imbalanced classes.

TABLE II
CONFUSION MATRIX FOR k CLASSES

		Predicted Class				
		Class 1	Class 2	...	Class k	Total
Actual Class	Class 1	a_{11}	a_{12}	$\cdots$	a_{1k}	$a_{1,+}$
	Class 2	a_{21}	a_{22}	$\cdots$	a_{2k}	$a_{2,+}$
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	Class k	a_{k1}	a_{k2}	$\cdots$	a_{kk}	$a_{k,+}$
	Total	$a_{+,1}$	$a_{+,2}$	$\cdots$	$a_{+,k}$	$n_{samples}$

For confusion matrices, the sum of the main diagonal means the total of correctly labelled samples by the model. Also, the value $a_{c,+}$ is the sum of all actual samples for class c , while $a_{+,c}$ is the sum of all samples assigned to class c and $n_{samples}$ is the total number of samples for validation.

From the confusion matrix (Tab II), we can observe that the True Positive (TP) values are $a_{i,i} \forall i \in [1, k]$, being the correctly labelled samples. The False Negative (FN) value for a class i is $a_{i,+} - a_{i,i}$, being the omitted samples. False Positive (FP) value is defined by $a_{+,i} - a_{i,i}$. Finally, True Negative (TN) value for class c is $\sum_{i=1, i \neq c}^k a_{i,i}$.

The Overall Accuracy for k classes is computed by Eq. 4, being related to the success rate of the model, out of all samples. On the other hand, the average Jaccard index was computed for all classes and it considered the intersection between predicted labels ($Pred$) and True labels ($True$), as shown in Eq 5; the average value was computed as general evaluation metric.

$$OA = \frac{\sum_{i=1}^k a_{i,i}}{n_{samples}} = \frac{TP + TN}{TP + FP + TN + FN} \quad (4)$$

$$IoU(Pred, True) = \frac{|Pred \cap True|}{|Pred \cup True|} \quad (5)$$

Then, for each class c , it is possible to compute Precision, Recall and Specificity, given by Eq.6, 7 and 8, respectively [1]. Precision is the ratio of the number of samples correctly belonging to that certain class c from the samples that were predicted to the class. Recall related to the ratio of the number of correctly labelled samples from samples actually from that class c . Specificity is the ratio between true negative class out of samples that are actually not from class c . Finally, F1-Score

¹The original model from PyTorch was trained using Imagenet dataset [15].

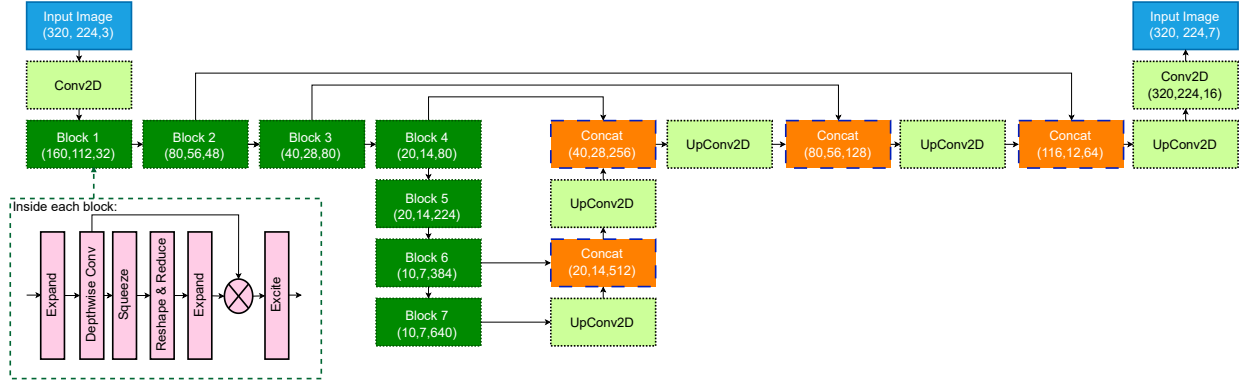


Fig. 3. Original Efficient-UNetB7 model framework for semantic segmentation. In each block, there is MBConv, as shown on the bottom left part of the image. Adapted from Baheti et al (2020) [8].

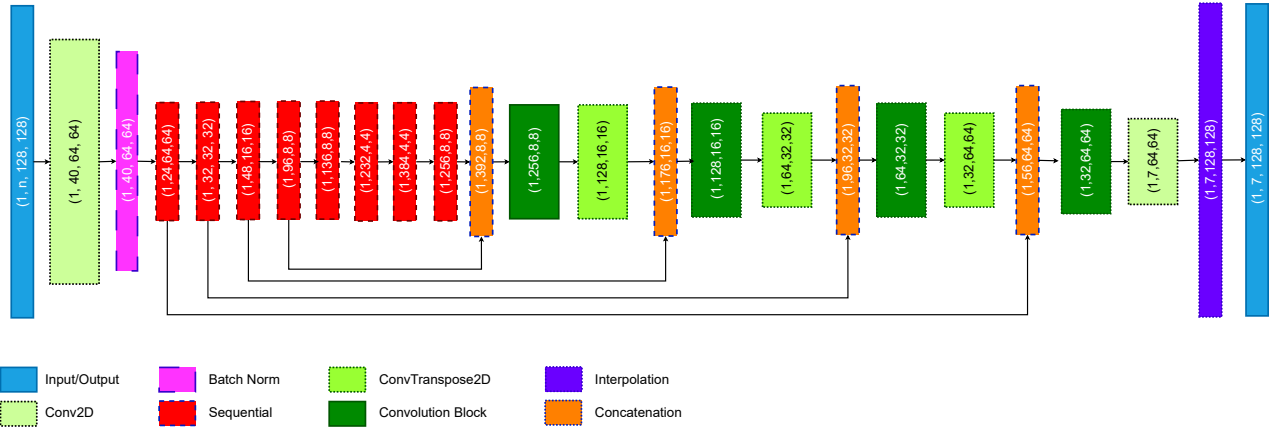


Fig. 4. EfficientUnetB3 semantic segmentation architecture adapted for n channels as input, with image size (128x128) and output for 7 classes, image size 128x128. This is a model summary considering a depth of 2. In this image, the output shapes of each layer are written inside each of them, respectively.

is the harmonic average of Precision and Recall for the class c . It is interesting to study each class separately because their representitivity can vary for each class in the dataset, as shown in Correa (2021) [20].

$$Precision = \frac{a_{c,c}}{a_{i,+}} = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{a_{c,c}}{a_{+,c}} = \frac{TP}{TP + FP} \quad (7)$$

$$Specificity = \frac{\sum_{i=1, i \neq n}^k a_{i,i}}{\sum_{i=1, i \neq n}^k a_{c,c} + a_{+,c} - a_{c,c}} = \frac{TN}{FP + TN} \quad (8)$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (9)$$

From these equations, it is possible to interpretate the data and its quality, as well as how the model can deal the these classes.

VI. RESULTS

The results are presented mostly by charts for the two setups, divided by the metrics observed: general metrics (Losses, OA, mIoU) are presented and discussed first and then metrics by class are presented later. For the model without pretraining, it stopped with 70 epochs, once there was no improvement of the loss for 20 epochs. The resulting general metrics for the model without pretraining is presented in Fig. 5. As can be seen, there is a difference of the loss between training and validation, indicating that the model overfitted for the training data, not being able to generalise as well as it should, with validation loss being stable around 0.75 and training loss stabilising around 0.65. It is proved by the same difference in OA and mIoU. Another point is the variation between OA and mIoU. One must observe that OA generalises for all samples in all classes, clearly not being well-suited for imbalanced classes. On contrair, mIoU balances all classes as it computed the average quality. In this sense, the model was not likely to well separate all classes.

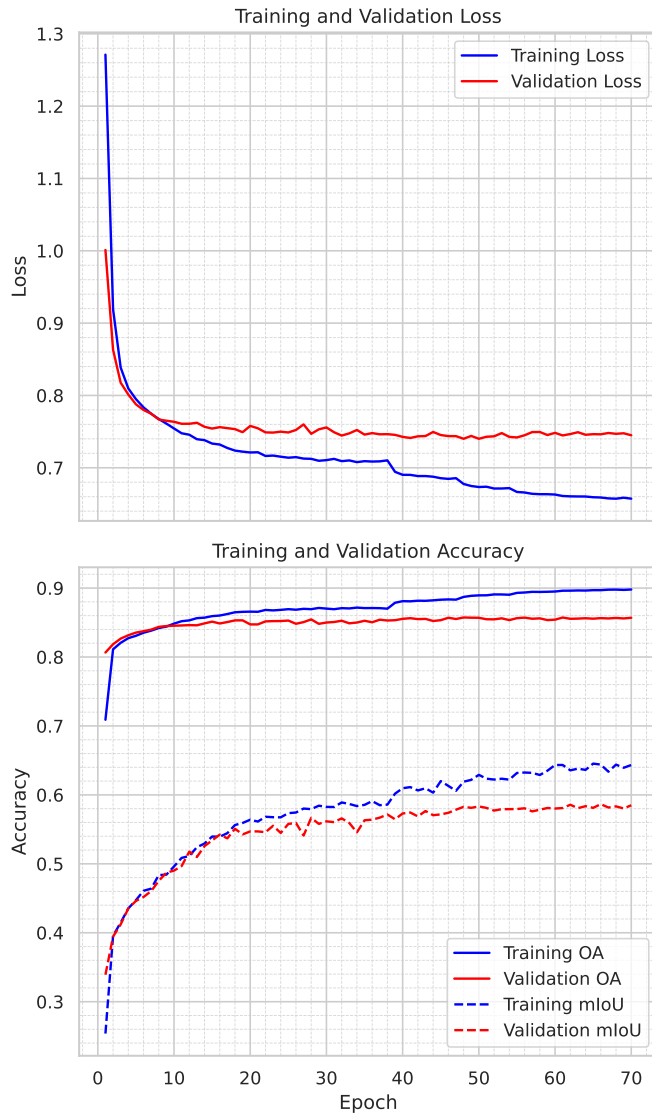


Fig. 5. General loss (left) and accuracy metrics (right) for model without pretraining. For the metric accuracies, Overall Accuracy (OA) is presented in continuous line, while mean intersection over union (mIoU) is presented in dashed lines. Red lines represent validation and blue lines represent training.

Now analysing the accuracy by class for the model without pretraining (Fig. 7), we can observe the imbalance; not only in the probable variation in the amount of samples per class², but in the metrics. In a nutshell, only three out of the seven classes were somehow distinguished: pasture, agriculture and arboreal. Proof of this relies on the Specificity of the remaining classes, that was set to 1.0 for all epochs. Apart from that, curiously, building class showed high precision in one single epoch, likely because there is small amount of buildings on the area and, in this specific epoch, the appeared in validation. Another interesting result is that pasture and arboreal classes seemed rather stable over the epochs, unlike agriculture, that showed more variation over the epochs, mostly in Precision.

²The dataset did not specify the number of samples per class.

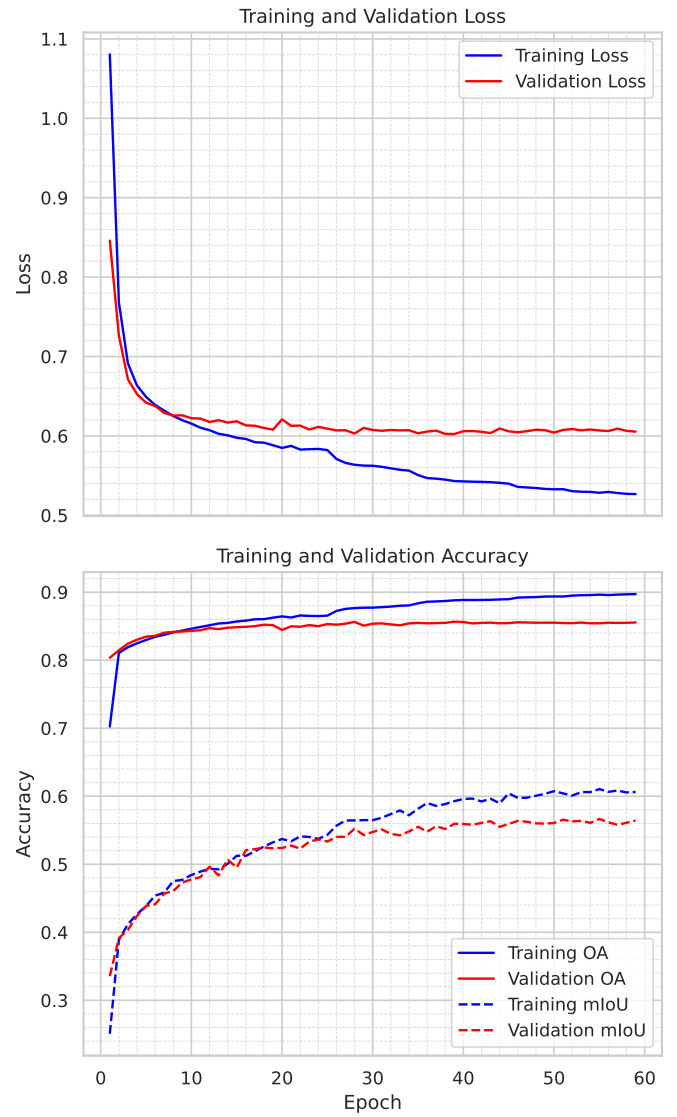


Fig. 6. General loss (left) and accuracy metrics (right) for model with pretraining. For the metric accuracies, Overall Accuracy (OA) is presented in continuous line, while mean intersection over union (mIoU) is presented in dashed lines. Red lines represent validation and blue lines represent training.

Now, moving on to the analysis of the model with pretrained weights, we can observe increase in accuracy for the losses and accuracy metrics (Fig. 6), though showing the same patterns as in the model without pretraining, showing a probable overfitting of the data. Though, this model showed Early Stopping at epoch 59, hence being faster to converge, which is a pro. Another point is regarding the losses, that this model was able to decrease final loss in about 0.1, in spite of the accuracies in general did not show this difference, when comparing model with and without pretraining data, which raises the question: so how is the model with pretraining better, with better loss if the accuracies did show subtle differences?

The answer to this question lies on Fig. 8. The first point is in Precision, that building class firstly showed up 3 more

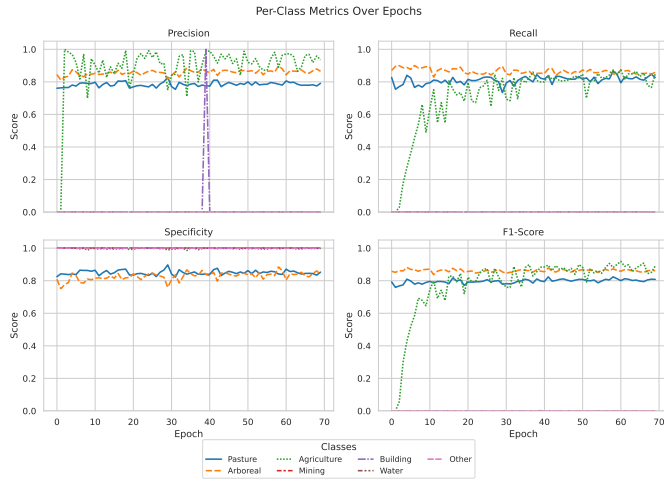


Fig. 7. Metrics by class for the model without pretraining. The top left metric is precision, followed by Recall on the top right. Specificity is on the bottom left and F1-Score is on bottom right of the image.

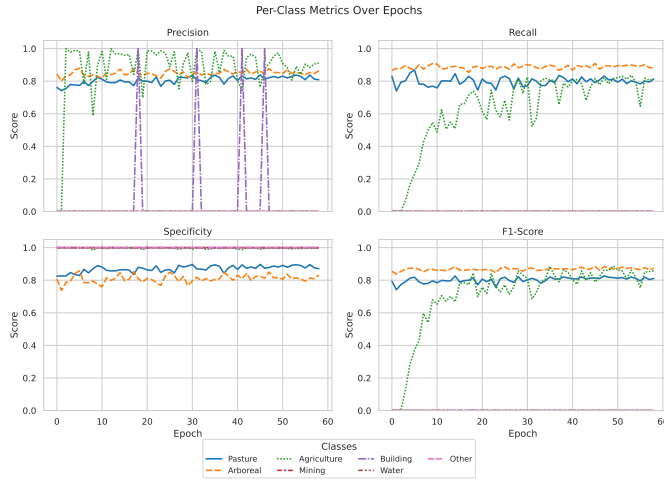


Fig. 8. Metrics by class for the model with pretraining. The top left metric is precision, followed by Recall on the top right. Specificity is on the bottom left and F1-Score is on bottom right of the image.

times, and pasture class showed some improvement. Arboreal class kept the same and agriculture presented higher variation. As for Recall, arboreal class did not drop, converging close to 0.9, while pasture and agriculture did not show significant changes. Specificity presented higher for pasture, showing that the model with pretraining was able to do little less confusion with this class than without it. Moreover, F1-score for agriculture and arboreal also showed some increase. An interesting fact is that, again, the other classes, Mining, Water, Other, were not able to show up in the charts, emphasising the class imbalance.

When comparing these results with those from Miranda et al (2025) [4] using the same dataset, we were able to obtain higher general accuracy, even though each study used different input features: our study added vegetation and soil indexes and excluded channels with band spatial resolution 10m, while

Miranda et al (2025) used all 12 channels plus the two SAR bands. On the other hand, when comparing accuracy by class, the other study presented some confusion between Mining, Water and Other classes, showing that the other bands may present some importance for creating the feature map in the model, despite they still being relatively low (≤ 0.5).

VII. CONCLUSION

In a nutshell, this study compared the same model considering pretraining and without considering it. General results showed that the model with pretrained weights converged with lesser amount of epochs and provided better losses. Interesting fact is that OA and mIoU did not show drastic differences, however, metrics by classes presented some better performance of the model using pretrained weights.

In general, using pretrained weights is encouraged and this study confirmed it. As recommendations, we consider an exploratory data analysis (EDA) of the dataset in a way to guarantee balanced classes so they can be best compared in future research. Another recommendation is to use pretrained data from the same satellite sensors, or even using self-supervised learning (SSL) to achieve better results.

VIII. CODE ACCESS

All used code was made available at <https://github.com/paeslema/projectsUFV>.

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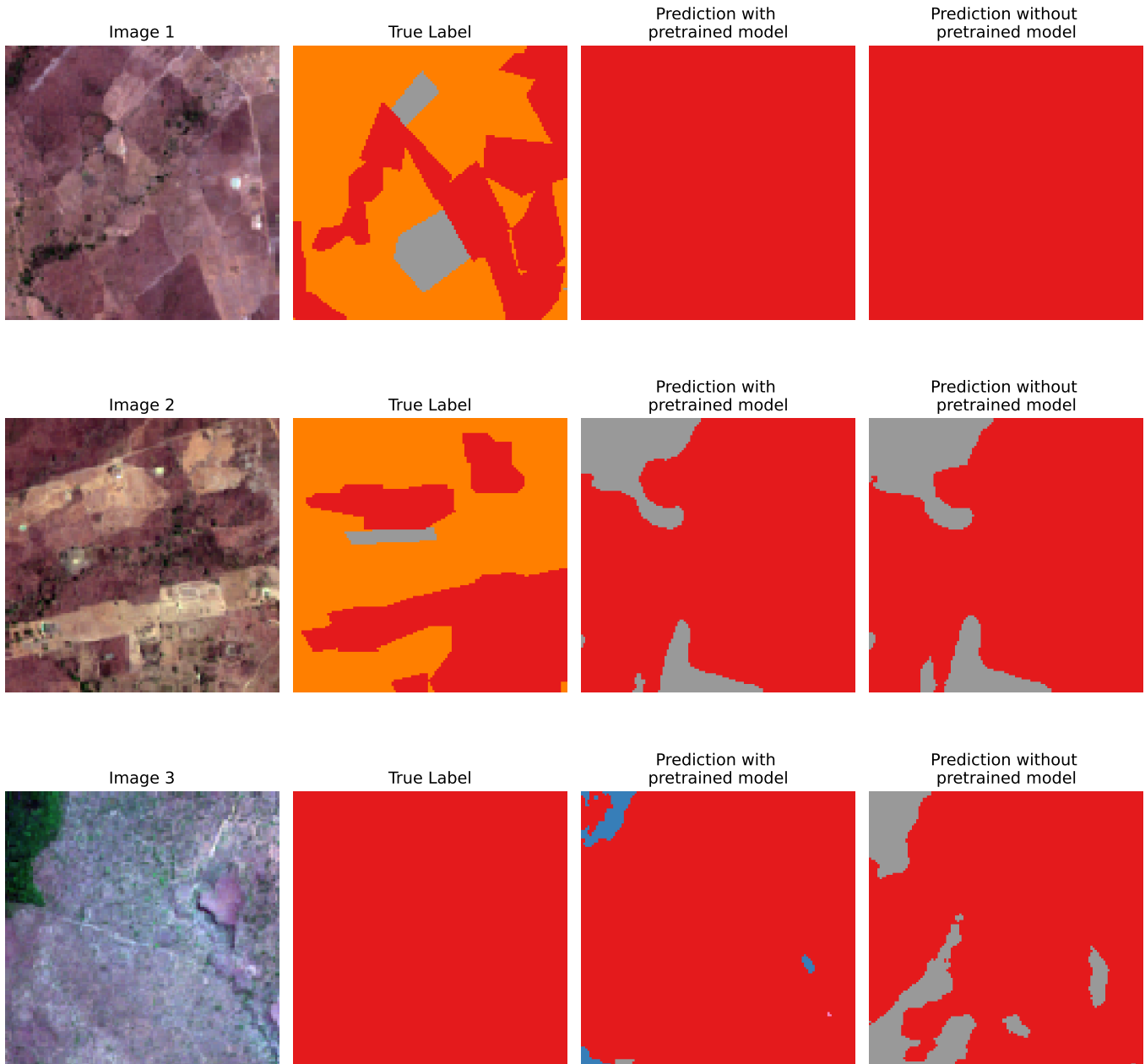


Fig. 9. Predictions for the models, showing the amount of confusion between the classes.

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